

SOC Home Lab Project

Inspired by [Eric Capuano's blog series: So You Want to Be a SOC Analyst?](#)

This project documents my hands-on journey into cybersecurity by replicating and expanding upon the practical lessons from Eric Capuano's SOC Analyst series. Through virtualization, adversary emulation, and detection engineering, I gained a solid foundation in SOC operations and modern blue teaming.

Part 1: Setting Up the Environment

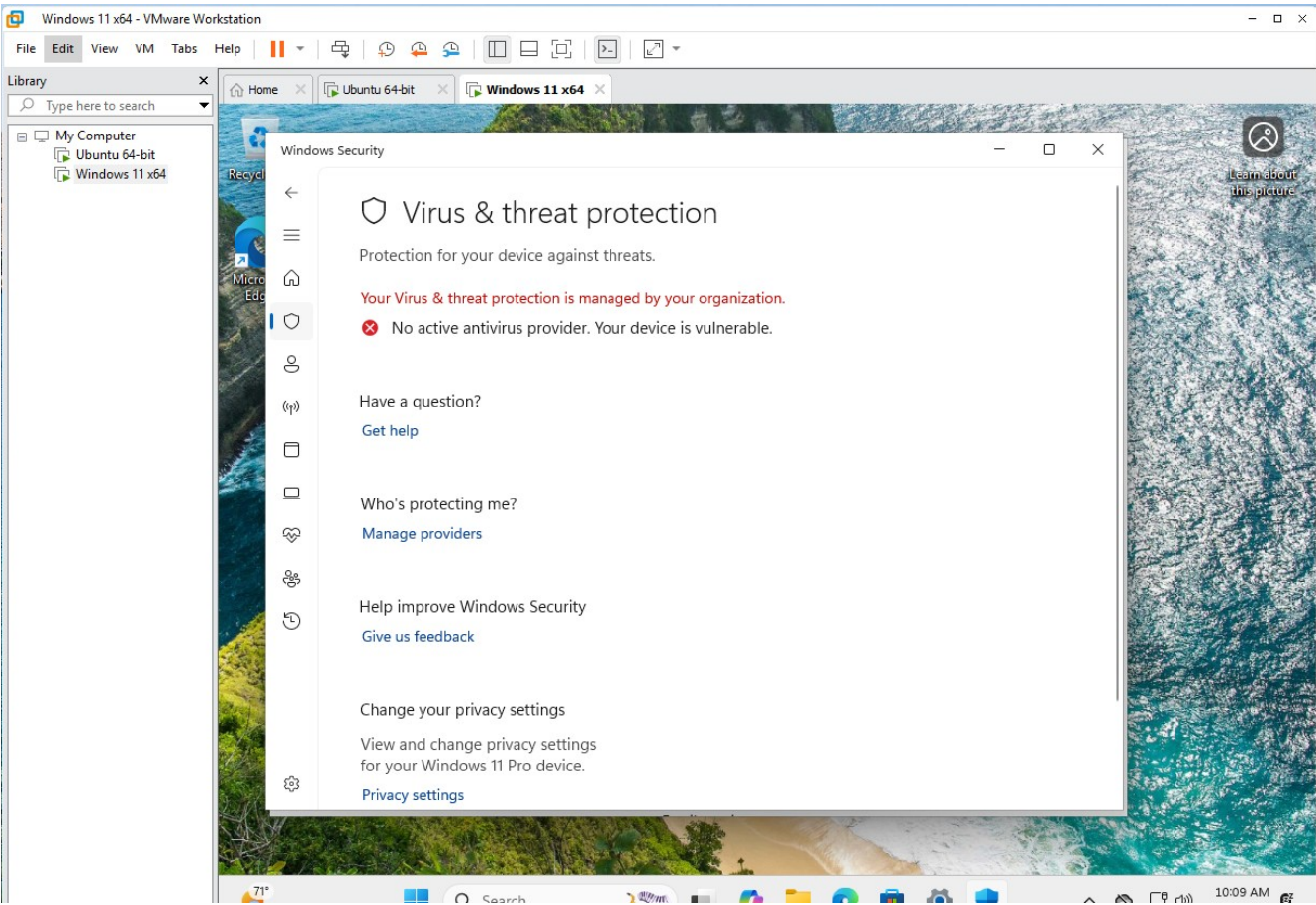
I set up a secure and isolated virtual environment for attack-defense simulations.

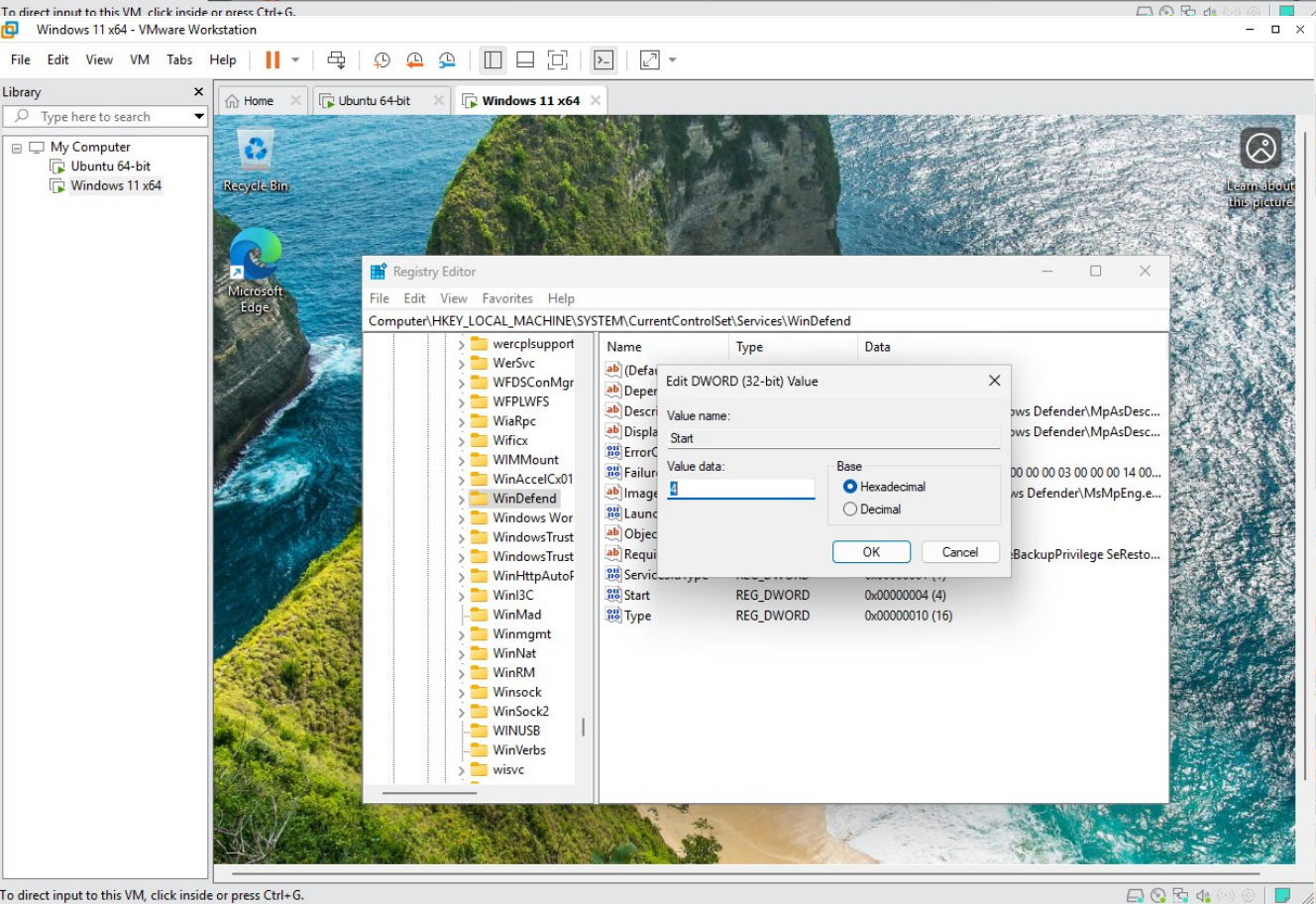
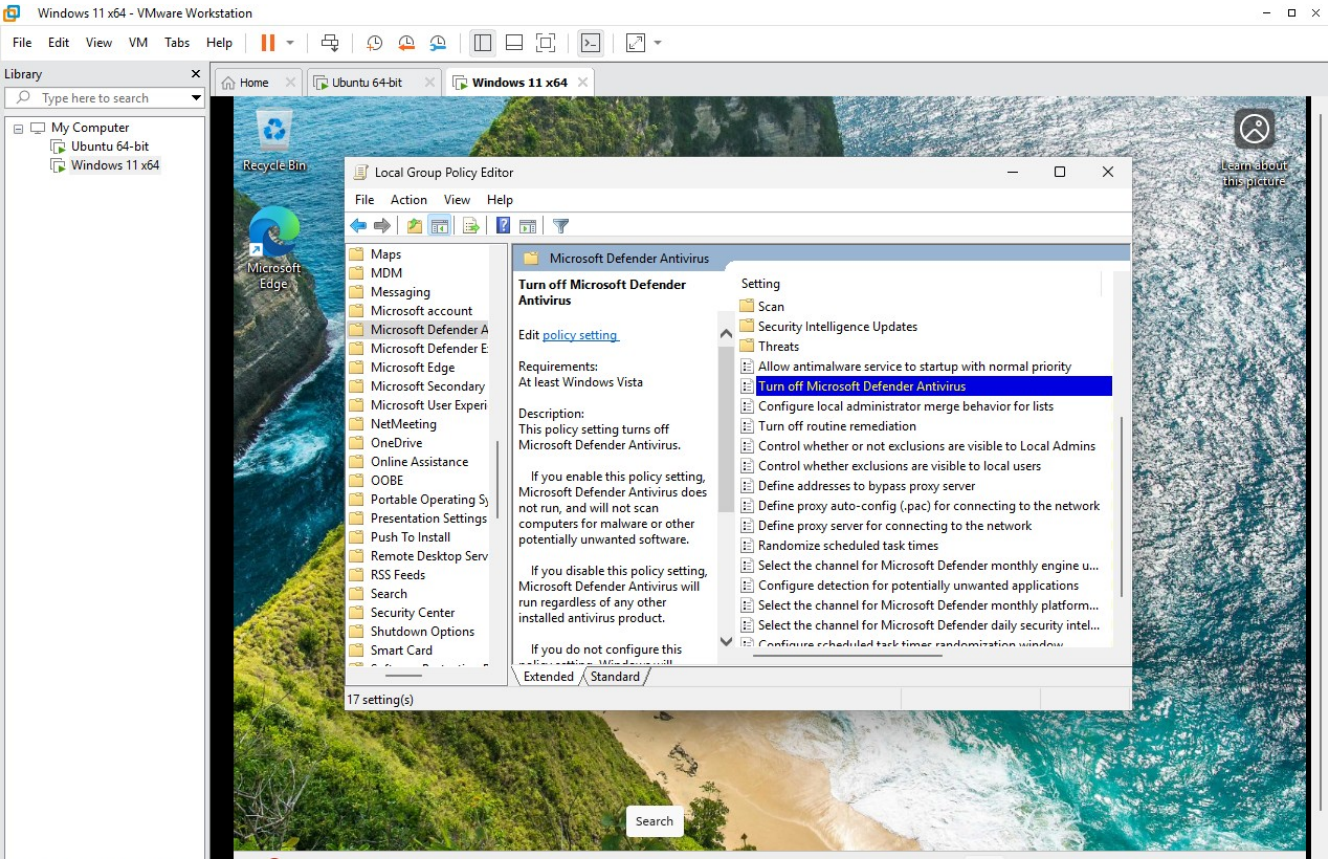
Virtualization Setup

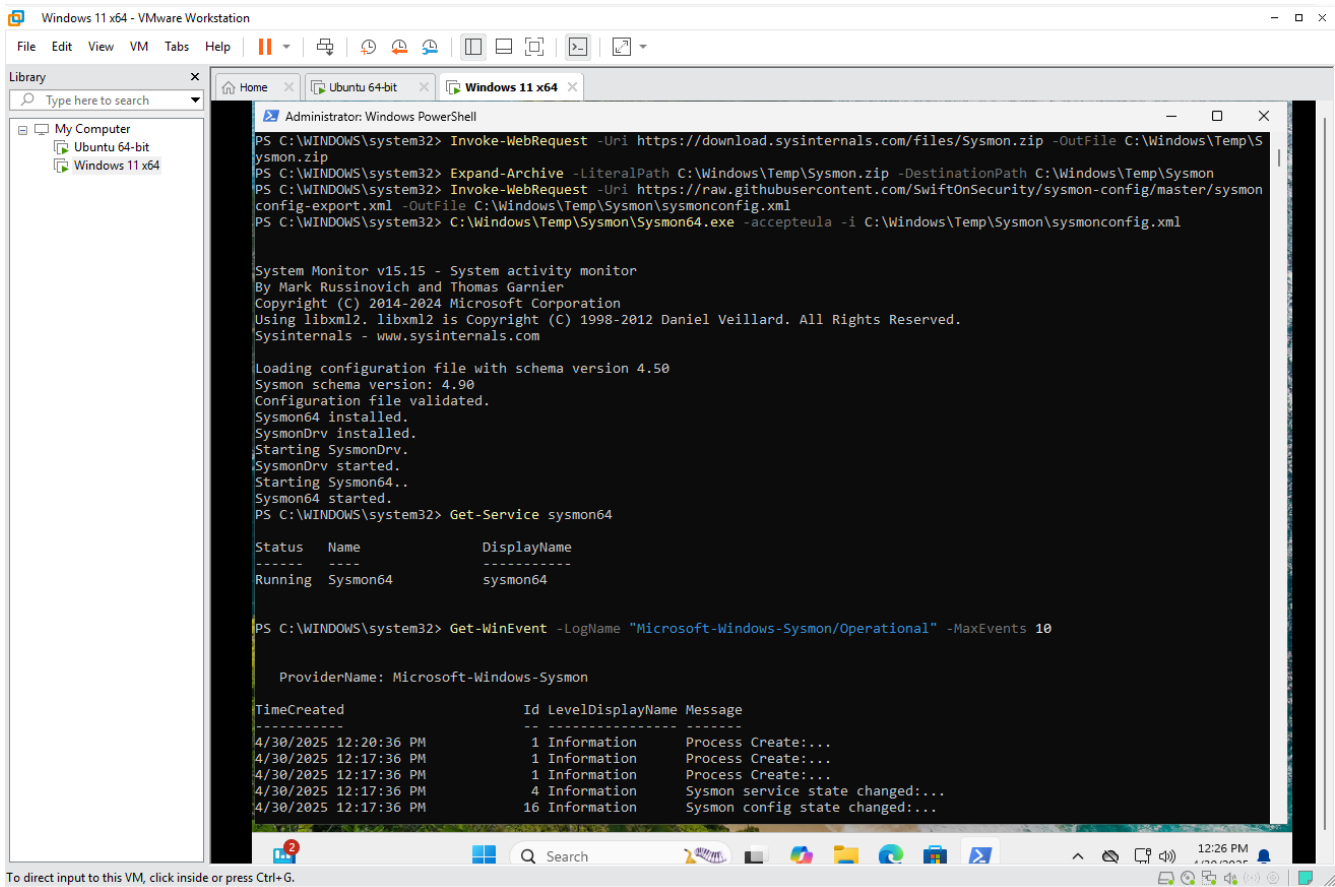
- Created two VMs:
 - **Linux (Attacker)** running Sliver, scanning tools
 - **Windows 11 (Victim)** configured for telemetry

Defender Hardening

- Disabled Windows Defender via Group Policy and registry
- Enabled Sysmon with a custom config for event collection







Part 2: Adversary Simulation

To understand how attackers operate, I used Sliver C2 to simulate intrusions.

C2 Server Setup

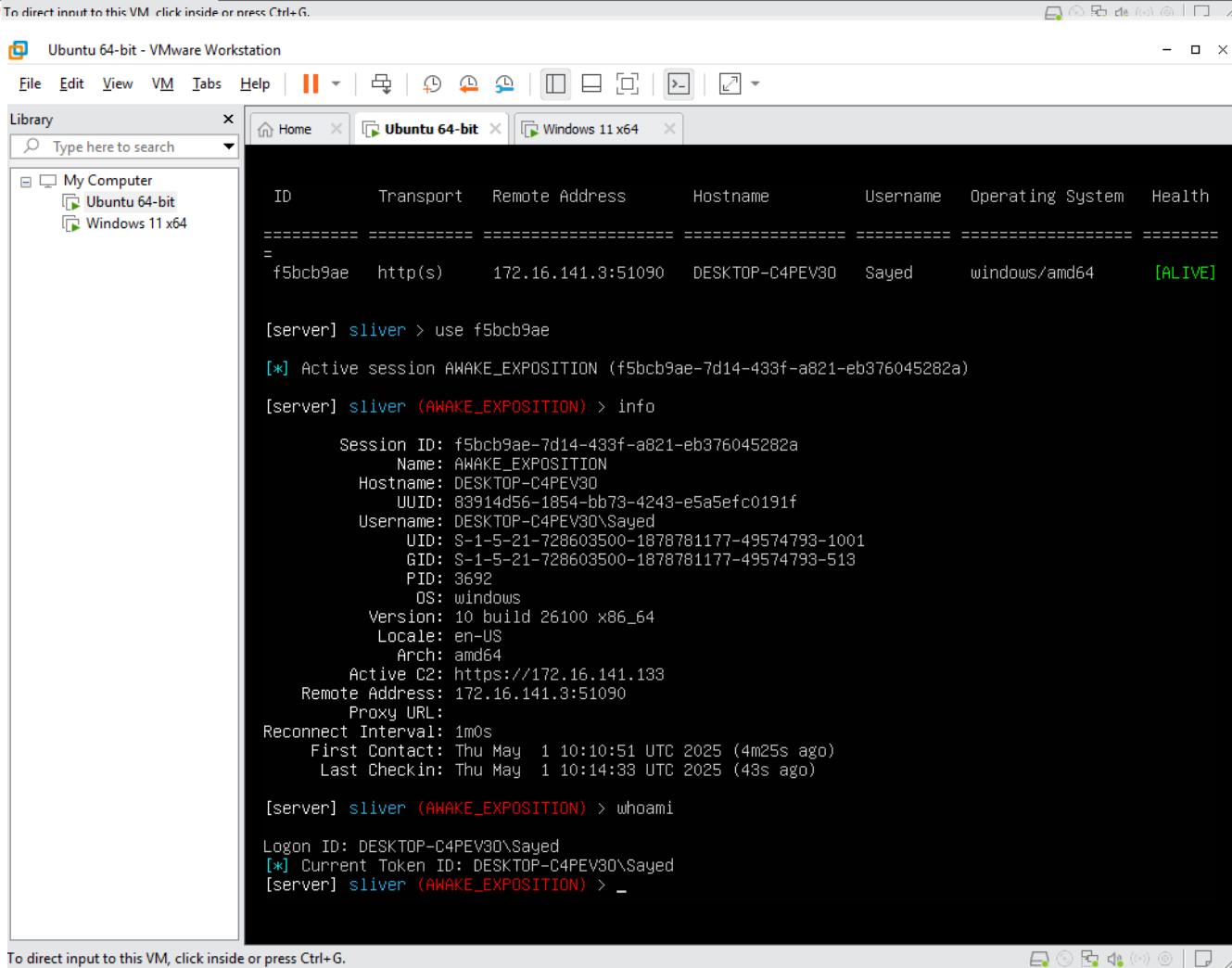
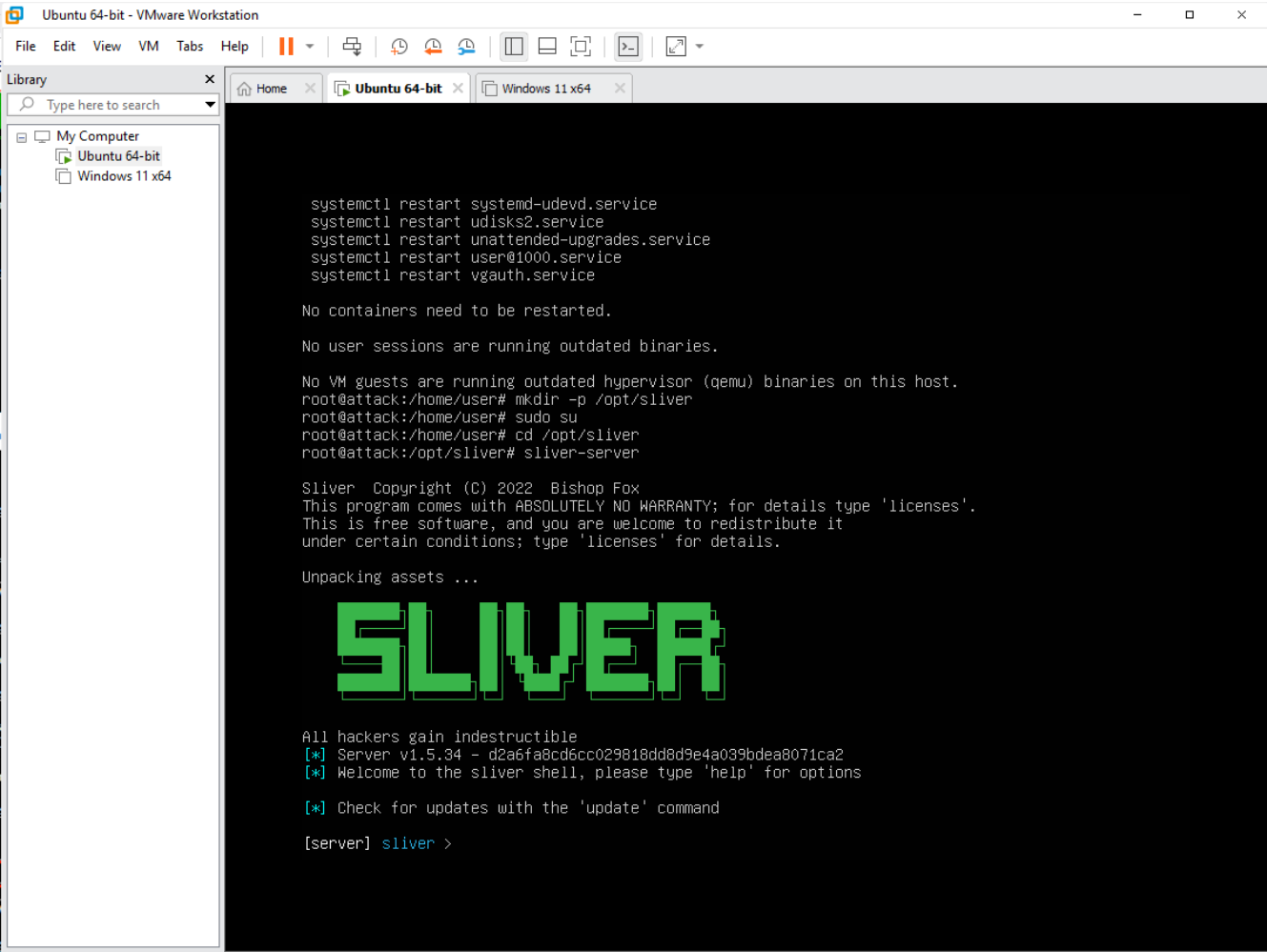
- Sliver installed on the Linux attacker VM
- Payload created and deployed to Windows VM

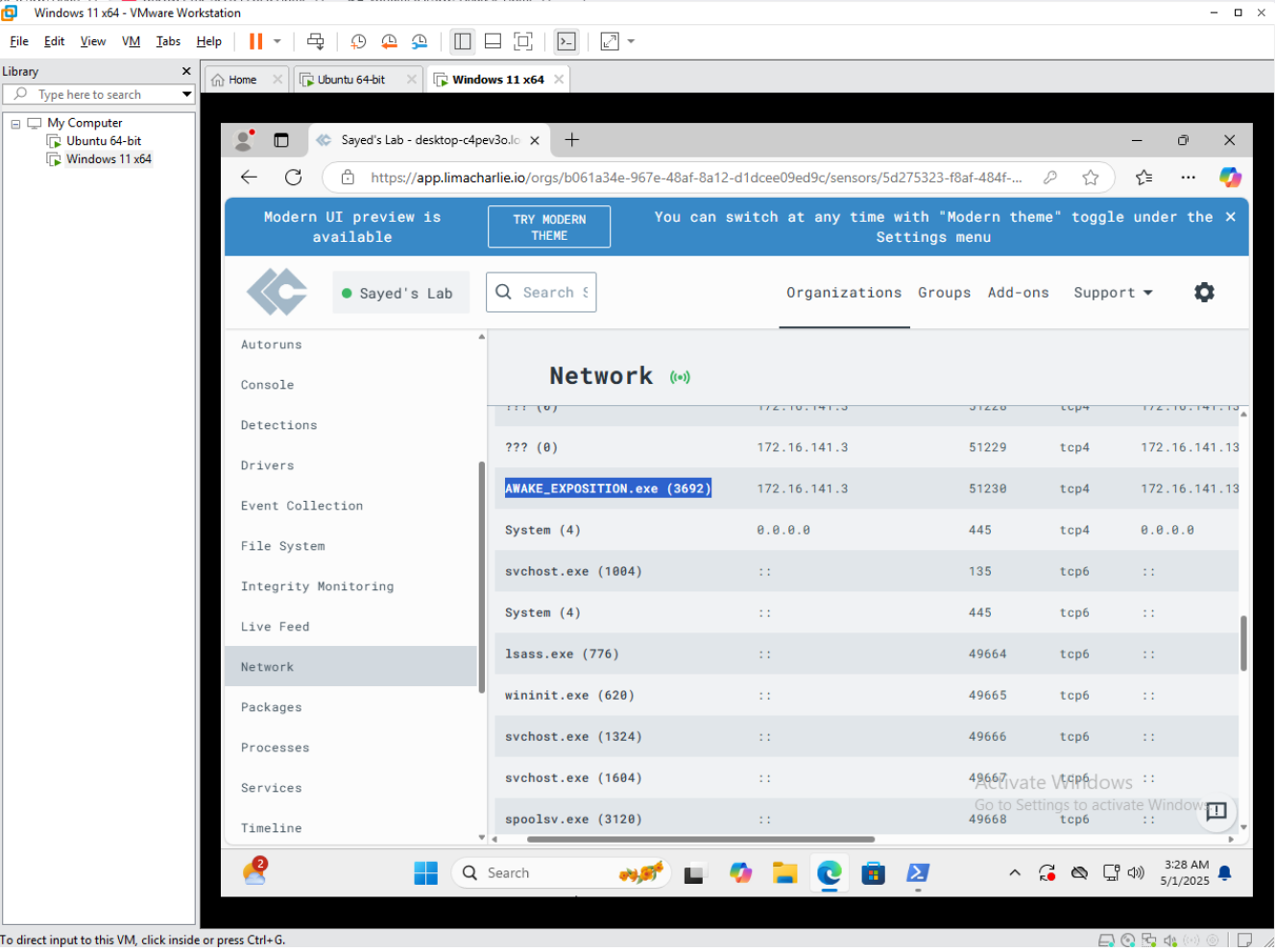
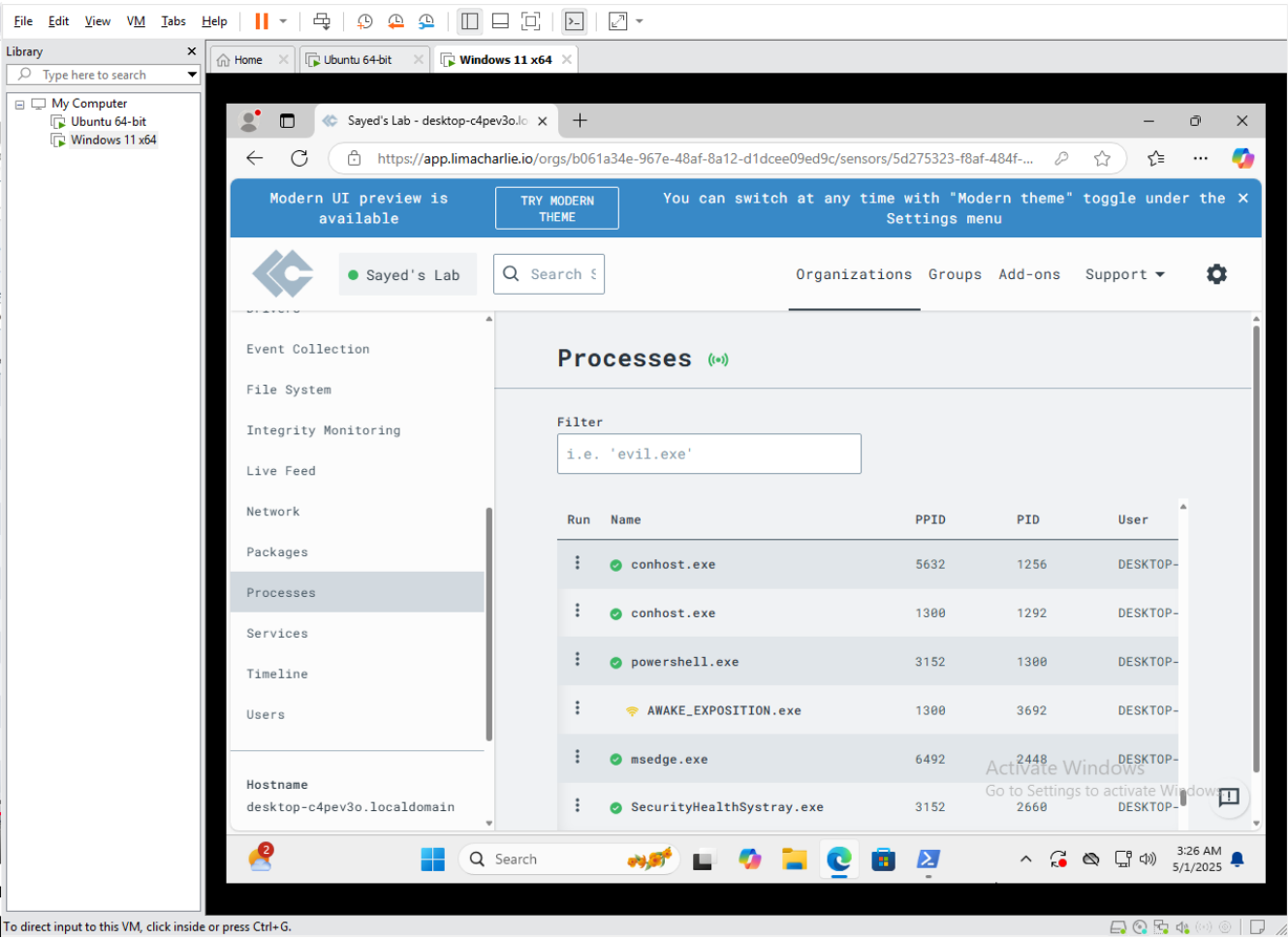
Payload Execution & Monitoring

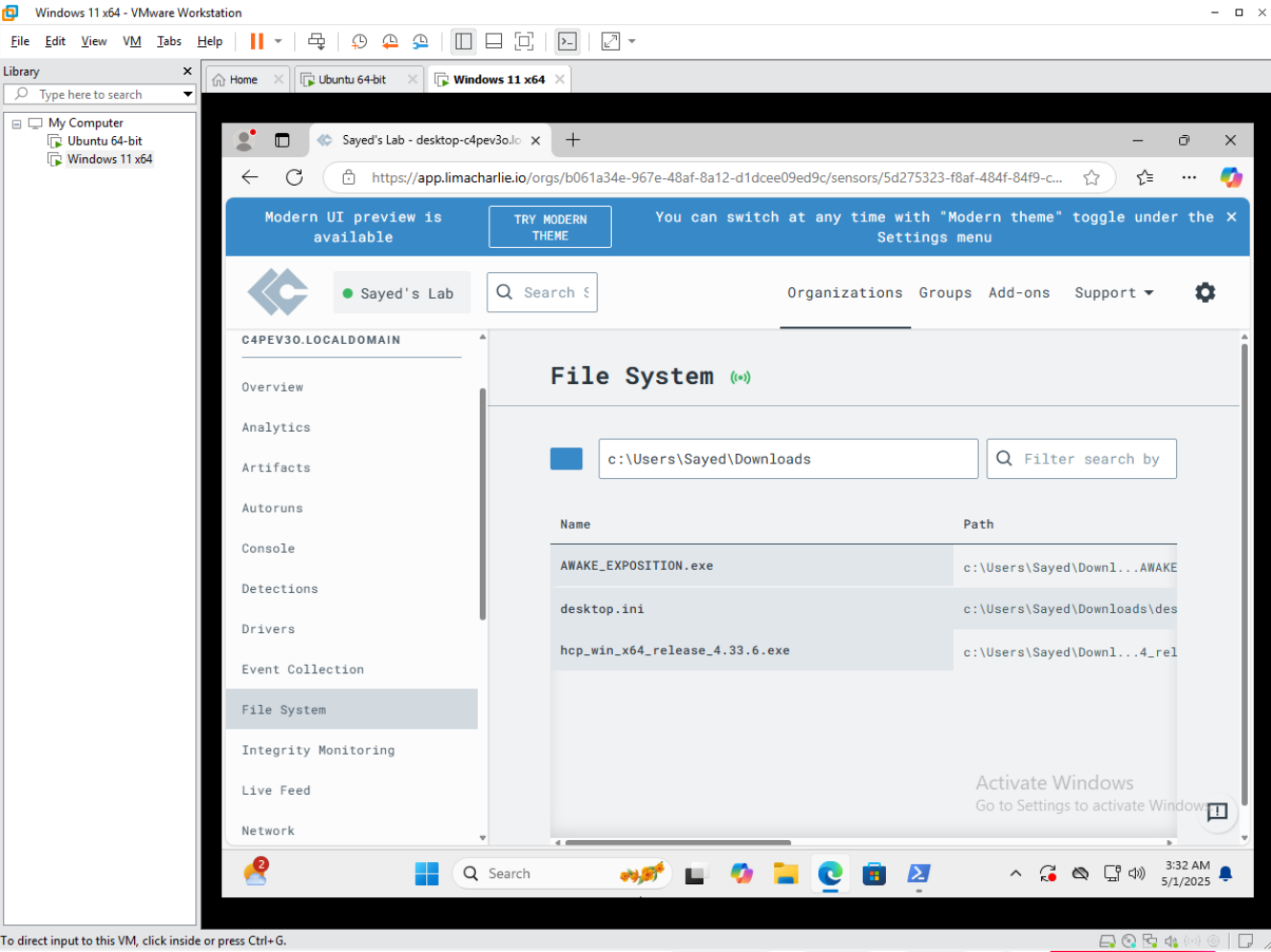
- Triggered the Sliver payload and established C2 comms
- Observed process creation, persistence, and command execution

EDR Telemetry

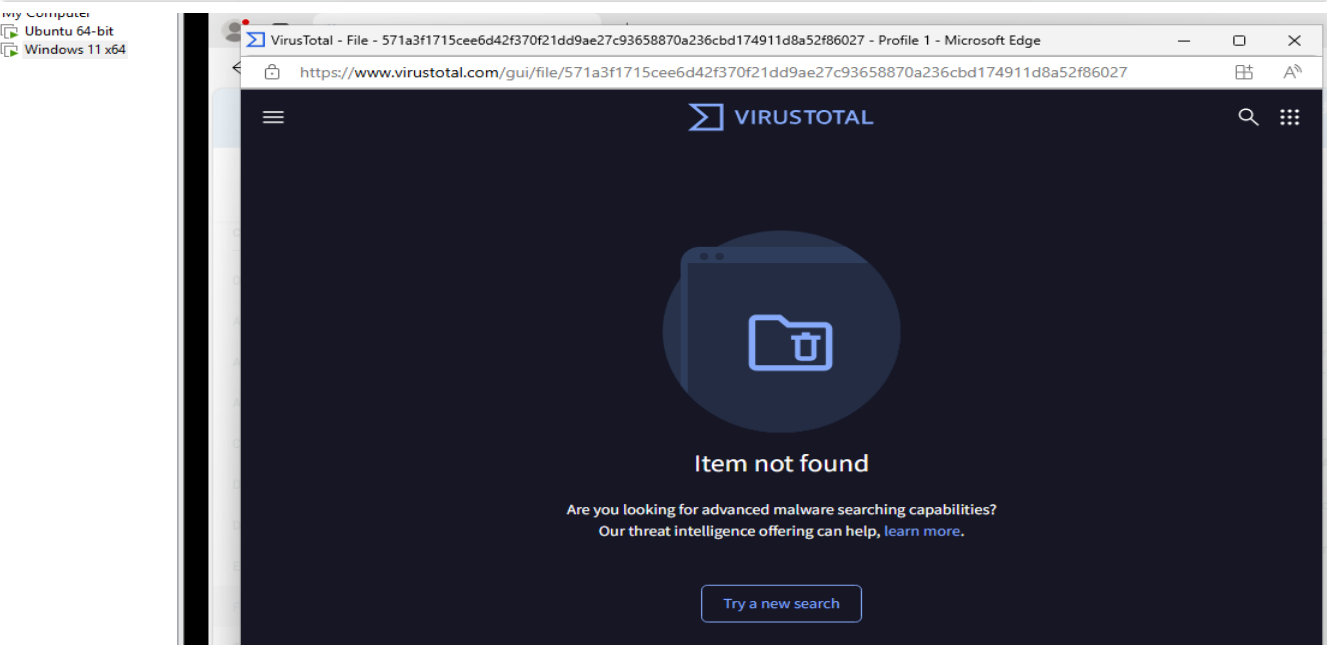
- Monitored the attack with LimaCharlie for logs and alerts







To direct input to this VM, click inside or press Ctrl+G.



Part 3: Crafting and Detecting Attacks

Focused on identifying and creating rules to catch attack techniques.

Credential Dumping

- Used procdump to dump lsass.exe
- Observed telemetry reflecting memory access and process injection

Detection Engineering

- Built detection rules in LimaCharlie based on telemetry and behavior patterns
- Flagged unusual access to sensitive processes

Untitled-2 [View Docs]

Detect

1 event: SENSITIVE_PROCESS_ACCESS

2 op: ends with

3 path: event/*/TARGET/FILE_PATH

4 value: lsass.exe

Expand

Op Reference

- and/or
- is
- exists
- contains
- starts with
- ends with
- is greater than
- is lower than
- matches
- string distance
- is :platform:
- is tagged
- lookup
- scope

Respond

1 - action: report

2 name: LSASS access

Comment

Add comment here. Comments are not included in detection content when an alert is triggered.

Save Rule

Discard Draft

Test Event

Match: 1 operations were evaluated with the following results:

- true => (ends with) ("event":"SENSITIVE_PROCESS_ACCESS", "op":"ends with", "path":"event/*/TARGET/FILE_PATH", "value":"lsass.exe")

Save New Rule

Name

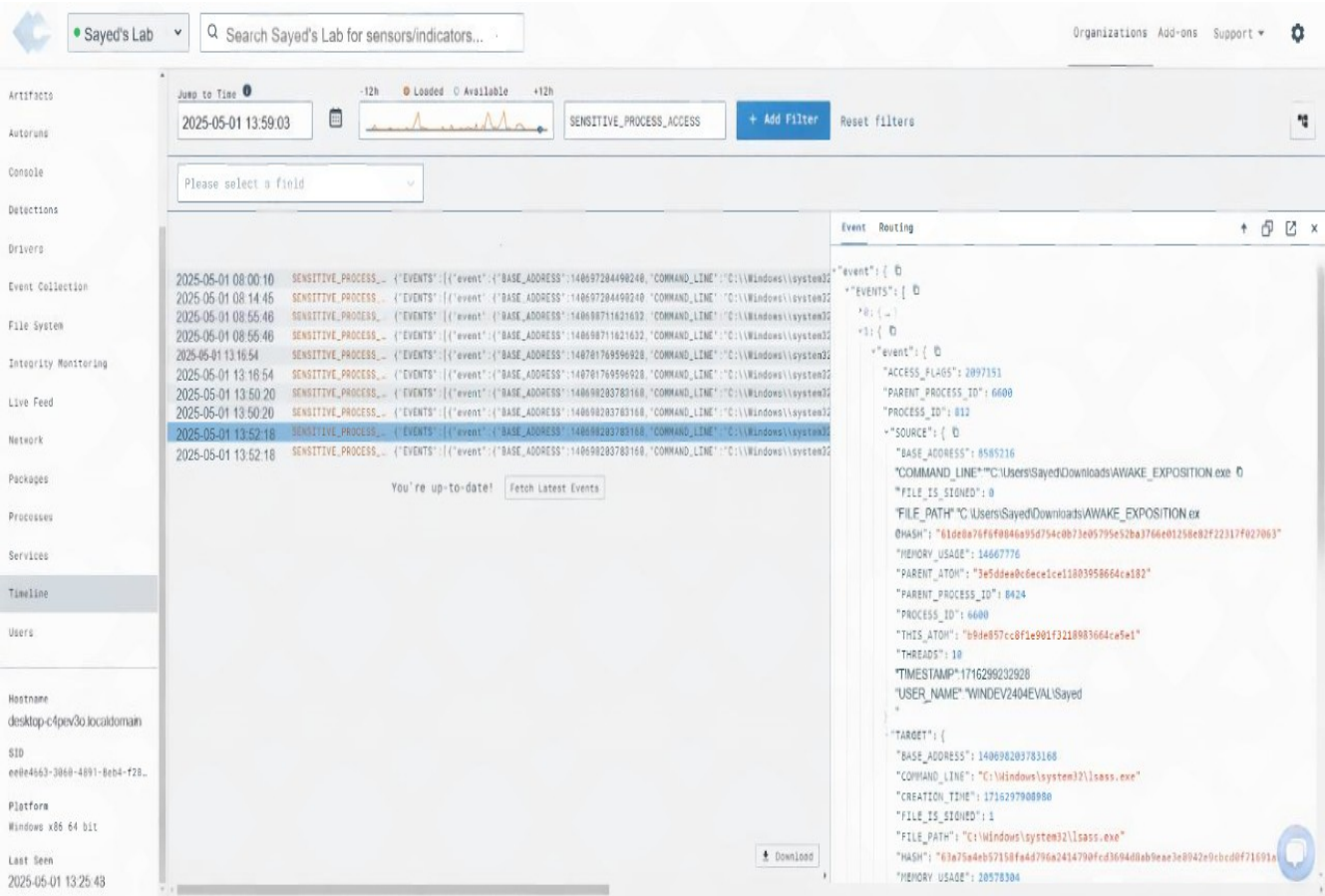
LSASS Accessed

Expiry Date (optional)

Select time...

Enabled

Save



Part 4: Blocking Attacks

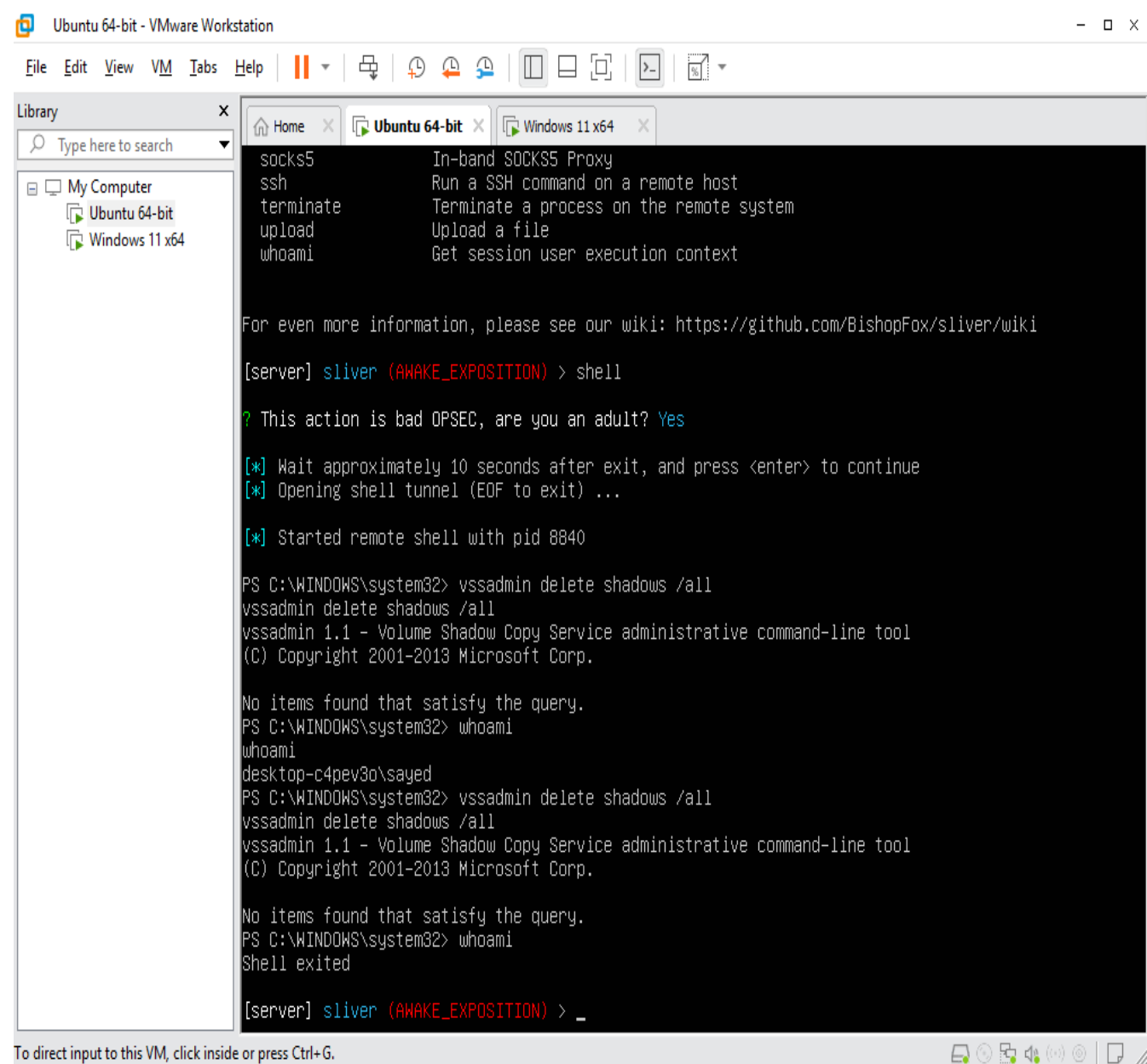
Developed and tested real-time response rules.

Ransomware Behavior Simulation

- Executed “vssadmin delete shadows /all” from the C2 shell

Blocking Rule

- Wrote a LimaCharlie rule to terminate the parent process when this command is detected



Windows 11 x64 - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

Ubuntu 64-bit

Windows 11 x64

Sayed's Lab - desktop-c4pev3o... x

https://app.limacharlie.io/orgs/b061a34e-967e-48af-8a12-d1dcee09ed9c/sensors/5d275323-f8af-484f-84f9-c9d...

Sayed's Lab

Search

Organizations Groups Add-ons Support

Event Collection

File System

Integrity Monitoring

Live Feed

Network

Packages

Processes

Services

Timeline

Users

Hostname

desktop-c4pev3o.localdomain

SID

5d275323-f8af-484f-84f9-c9d...

Date

Range

-12h

Loaded

Available +12h

2025-05-01 14:34:29

Quick Sea

Date	Time	Event	Priority
2025-05-01	14:25:45	NEW_PROCESS	Pr
2025-05-01	14:25:45	THREAD_INJECTION	PI
2025-05-01	14:25:45	TERMINATE_PROCESS	PI
2025-05-01	14:25:45	TERMINATE_PROCESS	PI
2025-05-01	14:25:46	NEW_PROCESS	Pr
2025-05-01	14:25:46	NEW_PROCESS	Pr
2025-05-01	14:25:46	NEW_PROCESS	Pr
2025-05-01	14:25:46	CODE_IDENTITY	Ha
2025-05-01	14:25:46	CODE_IDENTITY	Ha
2025-05-01	14:25:47	CODE_IDENTITY	Ha
2025-05-01	14:25:47	CODE_IDENTITY	Ha
2025-05-01	14:25:47	CODE_IDENTITY	Ha
2025-05-01	14:25:47	TERMINATE_PROCESS	PI
2025-05-01	14:25:47	CODE_IDENTITY	Ha
2025-05-01	14:25:47	NEW_PROCESS	Pr
2025-05-01	14:25:57	CODE_IDENTITY	Ha
2025-05-01	14:25:58	CODE_IDENTITY	Ha
2025-05-01	14:25:59	TERMINATE_PROCESS	PI
2025-05-01	14:26:02	NETWORK_CONNECTIONS	Cc
2025-05-01	14:26:02	NETWORK_C	Download
2025-05-01	14:26:02	NETWORK_C	Download

Event Routing

Event

Routing

event: {

COMMAND_LINE:

"C:\WINDOWS\system32\vssadmin.exe" delete shadows /all

FILE_IS_SIGNED: 1

FILE_PATH:

"C:\WINDOWS\system32\vssadmin.exe"

HASH:

"1b1d61fa237726bfd368fe601b059dd2d18124063d3017b22fde9eb79d15bd3"

PARENT: {

BASE_ADDRESS: 140700978053120

COMMAND_LINE:

"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -NoExit -Command [Console]::OutputEncoding=[Text.UTF8Encoding]::UTF8"

FILE_IS_SIGNED:

FILE_PATH:

To direct input to this VM, click inside or press Ctrl+G.

Windows 11 x64 - VMware Workstation

File Edit View VM Tabs Help

Library

My Computer

Ubuntu 64-bit

Windows 11 x64

Sayed's Lab - desktop-c4pev3o... x

https://app.limacharlie.io/orgs/b061a34e-967e-48af-8a12-d1dcee09ed9c/dr-general/vss_deletion_kill_it

Sayed's Lab

Search

Organizations Groups Add-ons Support

Back to Sayed's Lab

D&R RULES

New Rule

Untitled-1

Vss_deletion_kill_it

vss_deletion_kill_it

History

Detect

1 event: NEW_PROCESS

2 op: and

3 rules:

4 - op: is

5 path: event/FILE_PATH

6 value: C:\WINDOWS\system32\vssadmin.exe

7 - op: is

8 path: event/COMMAND_LINE

9 value: "C:\WINDOWS\system32\vssadmin.exe" delete shadows /all

10 - op: is

11 path: routing/hostname

12 value: desktop-c4pev3o.localdomain

13

Response

1 - action: report

2 name: vss_deletion_kill_it

3 - action: task

4 command:

5 - deny_tree

6 - <<routing/parent>>

7

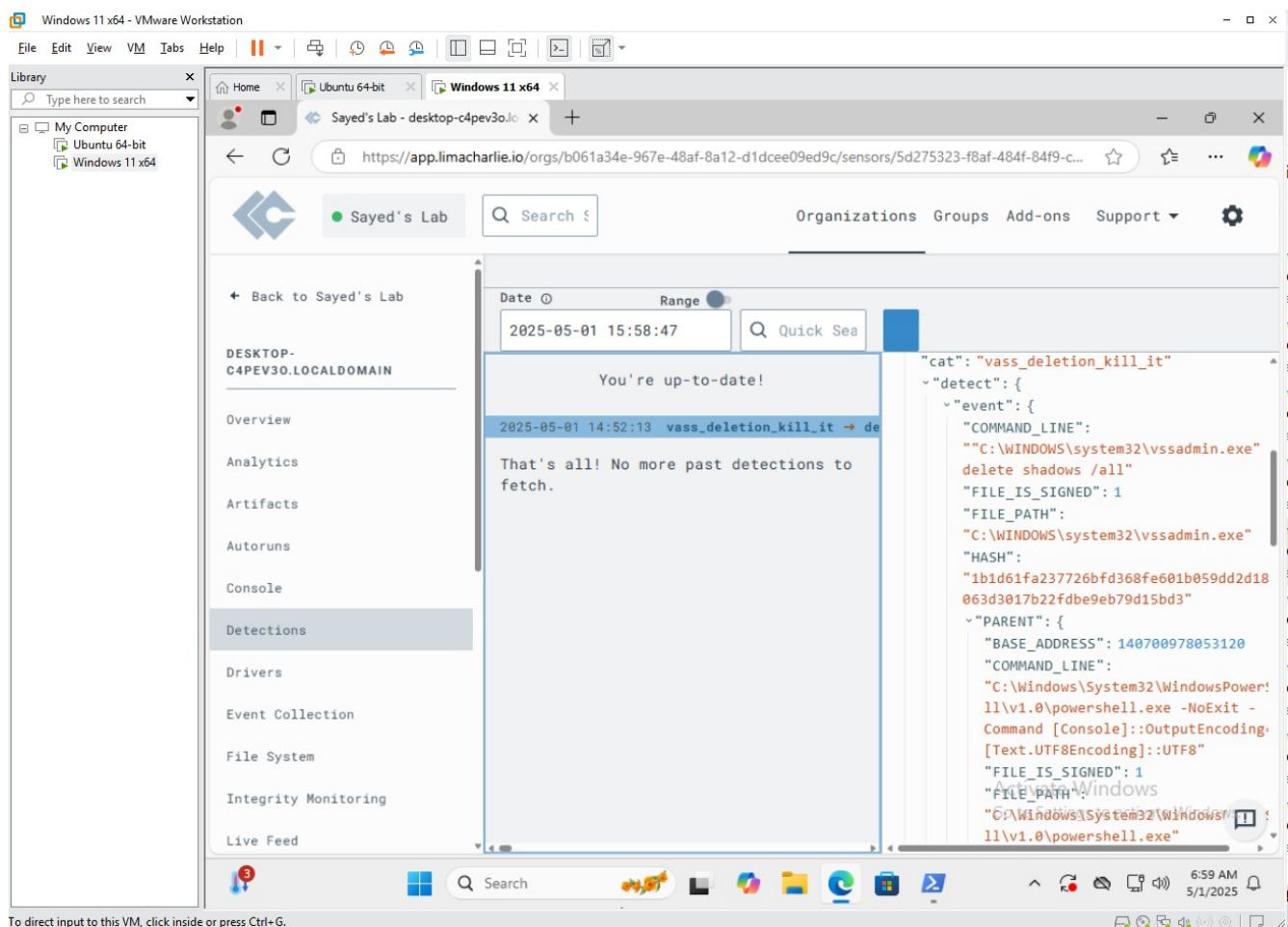
Op Reference

- and/or
- is
- exists
- contains
- starts with
- ends with
- is greater than
- is lower than
- matches
- string distance
- is :platform:
- is tagged
- lookup
- scope

Activate Windows

Go to Settings to activate Windows

To direct input to this VM, click inside or press Ctrl+G.



Part 5: Tuning False Positives

Although I did not do the hands-on exercise here, I studied the accompanying materials.

Key Takeaways

- Reviewed how to tune detection logic to reduce false positives
- Learned to balance detection breadth with noise reduction
- Understood rule conditions, exceptions, and filters

Part 6: Advanced Detection with YARA

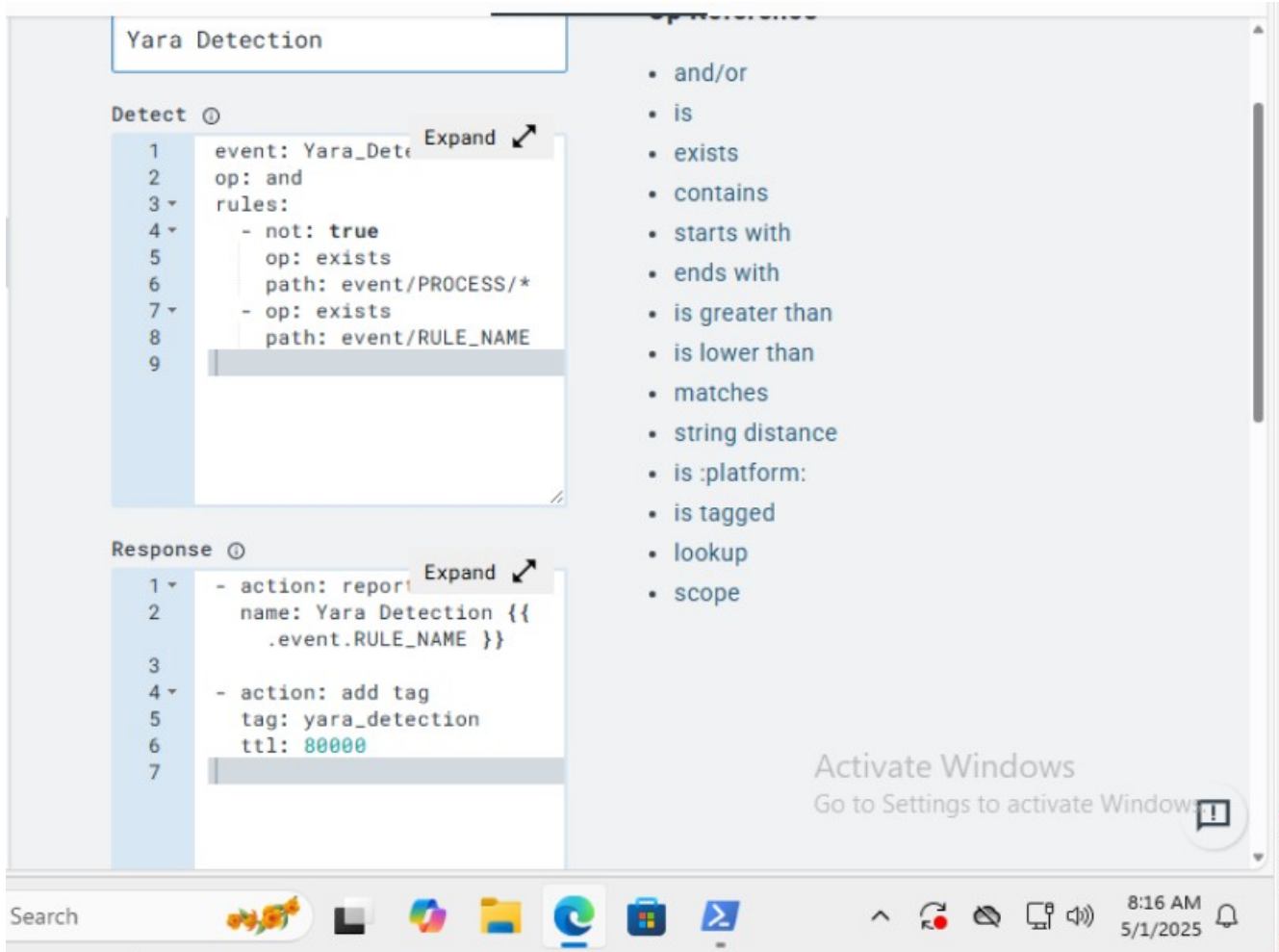
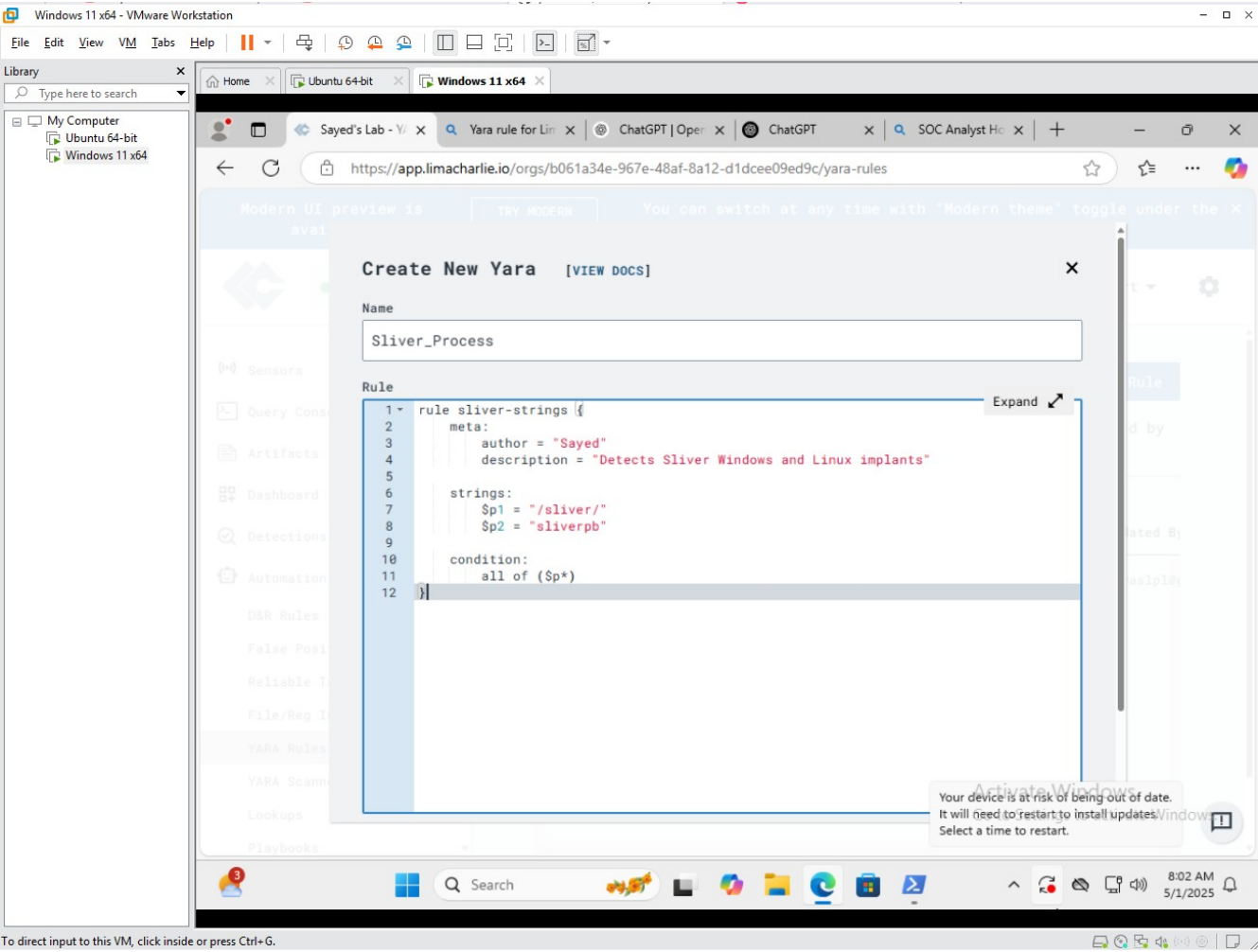
Advanced behavioral and signature-based detection with YARA rules.

Writing & Testing YARA

- Created rules targeting specific malicious patterns
- Manually tested them against simulated threats

Automation

- Automated YARA scans in the Downloads directory
- Scanned every new file or executable launched from it



Search

Organizations Groups Add-ons Support

Yara Detection in Memory

Detect

- event: Yara_Detection
- op: and
- rules:
 - op: exists
path: event/RULE_NAME
 - op: exists
path: event/PROCESS/*

Response

- action: report
name: Yara Detection in Memory {{ .event.RULE_NAME }}
- action: add tag
tag: yara_detection_memory
ttl: 80000

and/or

is

exists

contains

starts with

ends with

is greater than

is lower than

matches

string distance

is :platform:

is tagged

lookup

scope

Activate Windows
Go to Settings to activate Windows

Q

Search

Organizations

Groups

Add-ons

Support

⚙️

Yara Scan Downloaded Exe

Detect

1

event: NEW_DOCUMENT

2

op: and

3

rules:

4

- op: starts with

5

path: event/FILE_PATH

6

value: C:\Users\

7

8

- op: contains

9

path: event/FILE_PATH

10

value: \Downloads\

11

12

- op: ends with

Expand

Response

1

- action: report

2

name: EXE dropped in

3

Downloads directory

4

- action: task

5

command: >-

6

yara_scan hive://yara

7

/sliver -f "{{ .event

8

.FILE_PATH }}"

9

investigation: Yara Scan

10

Exe

11

suppression:

12

is_global: false

Expand

and/or

is

exists

contains

starts with

ends with

is greater than

is lower than

matches

string distance

is :platform:

is tagged

lookup

scope

Activate Windows

Go to Settings to activate Windows

!

Q

Search

Organizations

Groups

Add-ons

Support

⚙️

Yara Scan Process Launched from

Detect

1

event: NEW_DOCUMENT

2

op: and

3

rules:

4

- op: starts with

5

path: event/FILE_PATH

6

value: C:\Users\

7

- op: contains

8

path: event/FILE_PATH

9

value: \Downloads\

Expand

Response

1

- action: report

2

name: Execution from

3

Downloads directory

4

- action: task

5

command: >-

6

yara_scan hive://yara

7

/sliver-process --pid

8

"{{ .event.PROCESS_ID

9

}}"

10

investigation: Yara Scan

11

Process

Expand

and/or

is

exists

contains

starts with

ends with

is greater than

is lower than

matches

string distance

is :platform:

is tagged

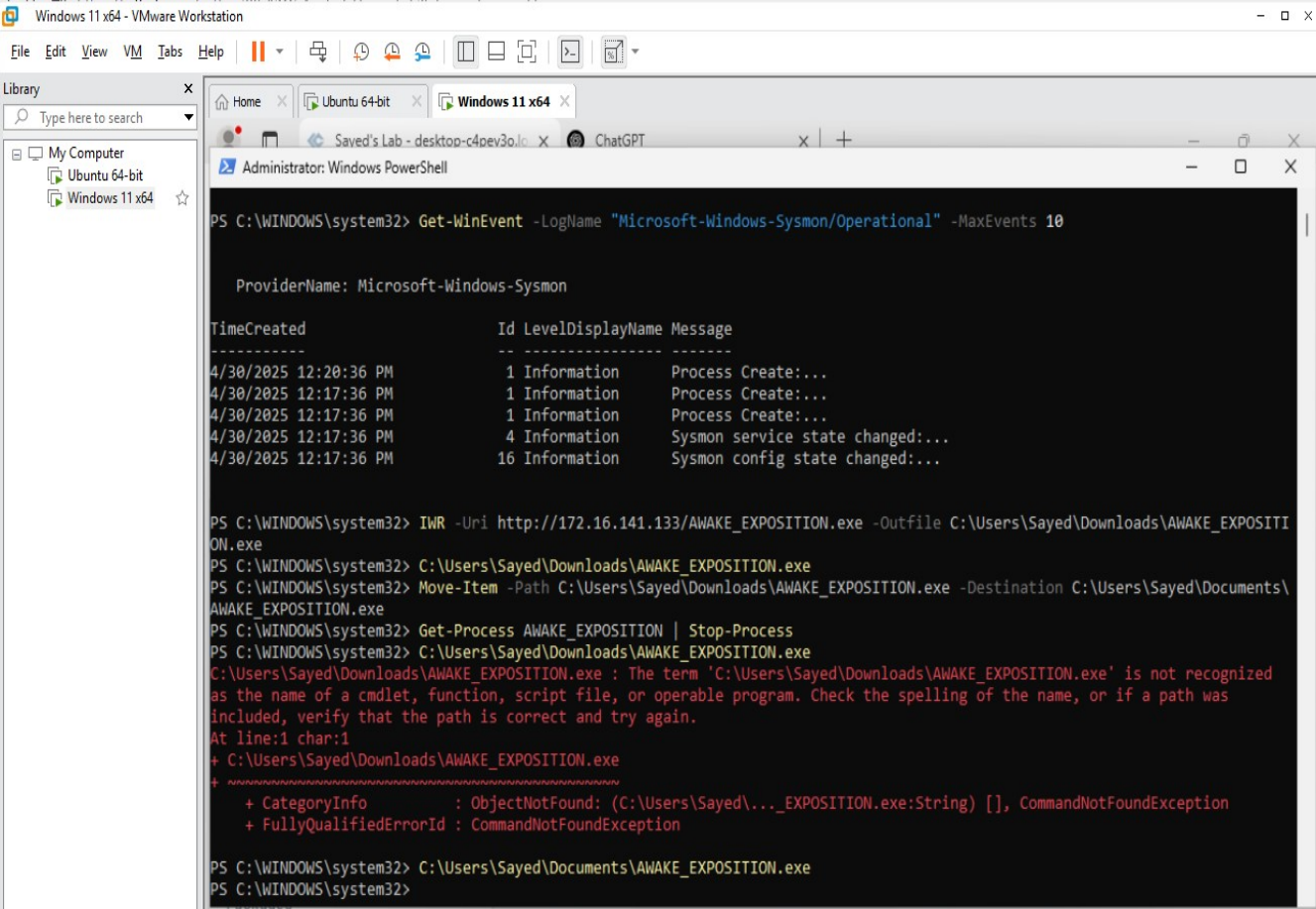
lookup

scope

Activate Windows

Go to Settings to activate Windows

!



Key Skills Gained

- Threat simulation and adversary emulation
- SOC-level detection engineering and EDR usage
- YARA-based detection automation
- C2 framework usage and endpoint telemetry analysis